

Data Science Analysis of The Daily Star Sports News Dataset

1. Introduction

Web scraping is a technique used to automatically extract data from websites by parsing their HTML structure. This project was performed to collect real-world sports news data for practicing Data Science workflows such as data collection, cleaning, and analysis. The Daily Star Sports section (thedailystar.net/sports) was used as the data source. A Python program built with Requests and BeautifulSoup collected news article titles, categories, dates, and URLs across multiple pages. The extracted records were stored using Pandas and exported into a structured CSV file named `daily_star_sports_news.csv` for further analysis.

2. Objectives

1. To collect sports news data from a live website using Python web scraping.
2. To create a structured CSV dataset from the scraped data.
3. To analyze the dataset using Pandas for basic statistics.
4. To identify missing values and duplicate records.
5. To analyze the distribution of news across sports categories.
6. To understand common data-quality issues in scraped datasets.

3. Tools and Technologies

Tool	Purpose
Python	Core programming language
Requests	Sending HTTP requests to the website
BeautifulSoup	Parsing HTML content
Pandas	Data processing and analysis
CSV	Storing the final dataset

4. Dataset Description

The dataset `daily_star_sports_news.csv` was generated by the scraping program described above. It contains 1,800 records and 4 columns: Title, Category, Date, and URL. Each row represents one sports news article collected from The Daily Star website.

Column	Description
Title	News article title/headline
Category	Sports category (e.g. Cricket, Football)
Date	Publication date, where available
URL	Link to the original news article

5. Data Analysis

Dataset Summary:

1. Total records: 1,800
2. Total columns: 4
3. Duplicate URLs: 0 (all URLs are unique)
4. Missing values — Title: 0, Category: 0, URL: 0, Date: 1,773

Category Analysis:

Category	Records	Percentage
Sports Special	992	55.1%
Football	357	19.8%
Cricket	338	18.8%
More Sports	54	3.0%
Tennis	42	2.3%
Sports	17	0.9%

Sports Special is by far the largest category, accounting for more than half of all collected records (55.1%), followed by Football (19.8%) and Cricket (18.8%). More Sports, Tennis, and Sports together make up only about 6.2% of the dataset, indicating that the website's content is heavily concentrated in the Sports Special, Football, and Cricket sections.

6. Data Quality Analysis

The most significant data-quality issue in this dataset is missing Date values: 1,773 out of 1,800 records (approximately 98.5%) have no date, with only 27 valid entries. This is because the scraper looked for a `<time>` HTML tag, and if absent, checked the parent element's text for the substring "2026" using code such as `time_tag = link.find("time")` and if "2026" in `parent_text: date = parent_text`. Since the actual webpage structure likely places publication dates elsewhere (e.g. inside the article page rather than the listing page), most Date fields could not be captured. In contrast, duplicate control was successful: the scraper used a `used_urls` set and a final `df.drop_duplicates(subset=["URL"])` step, and the dataset confirms 0 duplicate URLs, showing that the de-duplication logic worked correctly.

7. Findings

1. The dataset contains 1,800 unique, non-duplicated records across 4 columns.
2. Sports Special is the dominant category, representing over half of all articles.
3. Football and Cricket together account for roughly 38.6% of the dataset.
4. No duplicate URLs were found, confirming effective de-duplication in the scraper.
5. The Date column has a severe missing-data problem (98.5% missing), caused by the limited date-extraction logic in the scraping code.

8. Conclusion

This project successfully demonstrated a complete basic Data Science workflow — Data Collection, Data Cleaning, Data Analysis, and Findings — using real data scraped from The Daily Star Sports section. The resulting dataset of 1,800 records was clean in terms of duplication but showed a major weakness in date extraction, with only 1.5% of records containing valid dates. Future improvements could include visiting each individual article page to extract accurate publication dates and adding more robust HTML parsing logic to handle different page layouts on the website.